

VERHNEE PENZINO, Russian Federation

WMO: 255380

Lat: 64.217N

Lon: 164.233E

Elev: 326

StdP: 97.48

Time Zone: 12.00 (E12)

Period: 86-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -49.1	(c) -47.2	(d) -50.9	(e) 0.0	(f) -48.7	(g) -49.1	(h) 0.0	(i) -47.2	(j) 7.9	(k) -25.3	(l) 6.8	(m) -23.1	(n) 1.0	(o) 350	(p) 0.811

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.7	(c) 24.5	(d) 15.2	(e) 22.2	(f) 14.1	(g) 20.0	(h) 13.1	(i) 16.2	(j) 22.6	(k) 15.0	(l) 20.9	(m) 13.7	(n) 19.0	(o) 1.8	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 13.2	(b) 9.8	(c) 18.8	(d) 12.1	(e) 9.1	(f) 17.7	(g) 11.0	(h) 8.5	(i) 16.0	(j) 46.6	(k) 22.5	(l) 43.1	(m) 21.1	(n) 39.6	(o) 19.1	(p) 19.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.6	6.4	5.4	-51.4	27.7	2.8	1.5	-53.5	28.8	-55.1	29.6	-56.7	30.4	-58.8	31.5
			-50.8	17.5	2.7	1.1	-52.8	18.3	-54.4	18.9	-55.9	19.5	-57.9	20.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.8	-31.4	-26.6	-21.7	-13.0	0.3	10.3	13.1	9.4	2.5	-9.7	-21.7	-29.6
	DBStd	17.44	9.46	10.91	8.64	6.68	4.19	3.81	3.25	3.51	4.13	7.90	9.42	10.53
	HDD10.0	7403	1282	1025	984	691	300	42	9	52	225	611	952	1229
	HDD18.3	10258	1541	1258	1242	941	558	243	165	277	474	869	1202	1487
	CDD10.0	190	0	0	0	0	0	50	106	34	1	0	0	0
	CDD18.3	3	0	0	0	0	0	1	2	0	0	0	0	0
	CDH23.3	86	0	0	0	0	0	23	53	9	0	0	0	0
	CDH26.7	5	0	0	0	0	0	2	3	1	0	0	0	0
Wind	WSAvg	1.6	1.6	1.7	1.7	1.8	1.6	1.7	1.6	1.5	1.6	1.4	1.7	1.6
Precipitation	PrecAvg	322	24	16	14	12	13	31	44	42	34	26	34	24
	PrecMax	533	71	42	43	46	39	83	117	101	117	58	167	57
	PrecMin	199	0	0	0	0	0	4	9	9	0	6	4	3
	PrecStd	83	19	11	12	9	11	22	23	19	28	12	29	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.3	-2.0	-1.6	2.1	14.1	26.1	27.8	24.7	15.6	6.3	0.0	-2.1
		MCWB	-4.3	-3.6	-2.4	-0.3	7.4	14.7	16.8	16.2	9.1	3.9	-0.7	-3.2
	2%	DB	-8.8	-3.9	-3.6	0.3	10.9	23.6	25.4	21.1	12.7	4.0	-2.0	-6.1
		MCWB	-9.7	-4.7	-4.8	-1.4	5.5	13.8	15.9	14.6	8.0	2.2	-3.1	-6.8
	5%	DB	-14.0	-6.2	-6.1	-1.0	8.5	21.3	23.1	18.5	10.7	2.3	-4.7	-10.2
		MCWB	-14.6	-7.0	-7.2	-2.5	3.9	12.8	14.9	13.1	7.2	0.9	-5.3	-10.7
	10%	DB	-18.0	-9.6	-8.7	-2.8	6.3	19.1	21.0	16.3	9.2	0.4	-7.7	-14.0
		MCWB	-18.5	-10.2	-9.6	-4.3	2.6	11.5	13.9	11.8	6.5	-0.8	-8.4	-14.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.4	-3.3	-2.3	0.1	7.5	16.3	18.2	16.9	9.8	4.2	-0.6	-3.3
		MCDB	-3.3	-2.5	-1.7	1.5	13.3	22.6	24.4	23.5	13.2	6.1	0.0	-2.3
	2%	WB	-9.7	-4.6	-4.7	-1.0	5.6	14.7	16.8	15.2	8.8	2.5	-3.1	-6.7
		MCDB	-8.8	-3.7	-3.7	0.4	10.6	21.9	23.6	20.3	11.5	3.6	-2.1	-6.0
	5%	WB	-14.5	-7.0	-7.0	-2.6	4.1	13.3	15.7	13.6	7.9	1.0	-5.3	-10.5
		MCDB	-13.9	-6.2	-6.2	-1.1	7.7	20.3	21.9	17.6	10.0	2.2	-4.6	-10.1
	10%	WB	-18.4	-10.3	-9.6	-4.3	2.9	12.0	14.6	12.3	6.8	-0.8	-8.3	-14.6
		MCDB	-18.0	-9.6	-8.7	-2.9	5.9	18.8	20.2	15.7	8.9	0.3	-7.7	-14.1
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.9	11.5	11.5	7.4	11.2	10.7	10.3	9.4	8.2	8.6	7.7
		MCDBR	8.3	6.7	7.0	7.9	10.9	16.0	16.2	15.3	12.3	7.0	9.0	8.2
		MCWBR	7.9	6.2	5.9	6.3	6.4	8.2	8.2	9.1	8.1	5.3	8.0	7.9
	5% WB	MCDBR	8.3	6.6	6.1	7.4	9.7	14.5	14.2	13.1	9.7	6.3	8.6	8.4
		MCWBR	8.0	6.2	5.5	6.0	5.9	7.9	8.0	8.0	7.0	4.7	8.0	8.2
Clear-Sky Solar Irradiance	taub	0.166	0.236	0.296	0.355	0.401	0.375	0.447	0.426	0.311	0.262	0.183	0.165	
	taud	1.961	2.153	2.066	1.928	1.935	2.319	2.127	2.138	2.453	2.211	1.936	1.923	
	Ebn at Noon	447	697	773	797	793	840	752	726	782	641	374	178	
	Edh at Noon	25	58	99	145	159	112	132	119	70	54	25	10	
All-Sky Solar Radiation	RadAvg	0.17	0.94	2.74	5.01	5.72	5.10	4.86	3.60	2.17	1.03	0.28	0.06	
	RadStd	0.02	0.09	0.17	0.20	0.42	0.56	0.48	0.43	0.25	0.17	0.03	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page